AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

MIS200-IS

Relational Database Management Systems

Final Examination / Version: B

2025-2026 / FALL

15 – January – 2026

Number of Exam Pages: 2 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
This is a **CLOSED** book examination. NO documents of any kind are allowed.
You **MUST** answer the questions using only the answer booklets provided
You **MUST** return the examination paper together with your answer booklets.
Do **NOT** write your answers in pencil except graphs/sketches
Students are **NOT** allowed to use any type of dictionary or mobile phones during the examin
Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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YOU HAVE TO ANSWER ALL THE QUESTIONS

Question 1 (45 marks)

a) Construct an E-R diagram for an online retail store that includes products and customers. Associate with each customer a history of their orders, including product details, order dates, and payment status (35 Marks).

b) Explain the difference between a weak and a strong entity set (10 Marks) .

Question 2 (35 marks)

Consider the following clinic appointment database, where the primary keys are underlined.

patient (patient_id, name, phone)
doctor (doctor_id, name, specialization)
appointment (appointment_id, patient_id, doctor_id, appointment_date)

a) Write DDL statements for creating all tables with constraints. Make any reasonable assumptions about data types and be sure to declare primary and foreign keys (15 marks).

b) Write an SQL query to retrieve appointments with doctor name 'Dr. Ahmed' (5 marks).

c) Write an SQL query to insert a new patient record (assume the values) (5 marks).

d) Write an SQL query to delete an appointment using appointment_id (assume the appointment_id is known) (5 marks).

e) Write an SQL query to remove the entire doctor's table from the database (5 marks).

Question 3 (20 marks)

a) Explain briefly four SQL data types with examples (10 marks):

b) Explain how entities are transformed into tables (10 marks)

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